

Why is the body cavity of a cockroach not a true coelom but a haemocoel?

Fat bodies

The haemocoel contains many large sized fat bodies. These bodies are also called **corpora adiposa**. These are similar to the liver of the vertebrates in certain functions. A fat body has many lobules containing four types of cells namely **trophocytes** (store food) **mycetocytes** (contain symbiotic bacteria), **oenocytes** (synthesise and store lipids), **urate cells** (store uric acid).

7.3 LOCOMOTION

Periplaneta has double mode of locomotion. 1-Running (cursorial locomotion) 2-Flying.

1. Running or cursorial locomotion

Cockroach is a cursorial insect and it can run on the ground with the help of legs. During walking or running cockroach forms two **tripods** with the legs. Each tripod is formed by foreleg and hind leg of one side and middle leg of the other side. During locomotion, the three legs of one tripod are kept on the ground and the other three legs of another tripod are carried forward. In this process, the foreleg and the hind leg of the tripod pull and push the body while the middle leg acts as a **pivot**. Thus cockroach moves using the two tripods alternately. The animal does not slip back as plantulae, pulvilli and claws provide grip. The claws and the arolium help in locomotion on **rough surfaces** whereas plantulae are useful on **smooth surfaces**.

How is each tripod of cockroach formed during locomotion?

2. Flying: Though cockroach is a cursorial insect, it can fly over short distances with the help of its wings. While flying the first pair of wings is stretched out at right angles to the body. The second pair of wings is moved up and down during flight. Wings are elevated by the contraction of dorsoventral muscles. Contraction of the dorso-longitudinal muscles and relaxation of the dorsoventral muscles depresses the wings.

7.4 DIGESTIVE SYSTEM OF PERIPLANETA

The digestive system of cockroach consists of an alimentary canal and the associated glands. The preoral cavity, surrounded by the mouth parts, is present in front of the mouth. The hypopharynx divides it into two chambers called **cibarium** (anterior) and **salivarium** (posterior).

I. Alimentary canal

The alimentary canal of cockroach is a long tube and is coiled at some places. It extends between the mouth and the anus. It is divided into three regions, namely, foregut